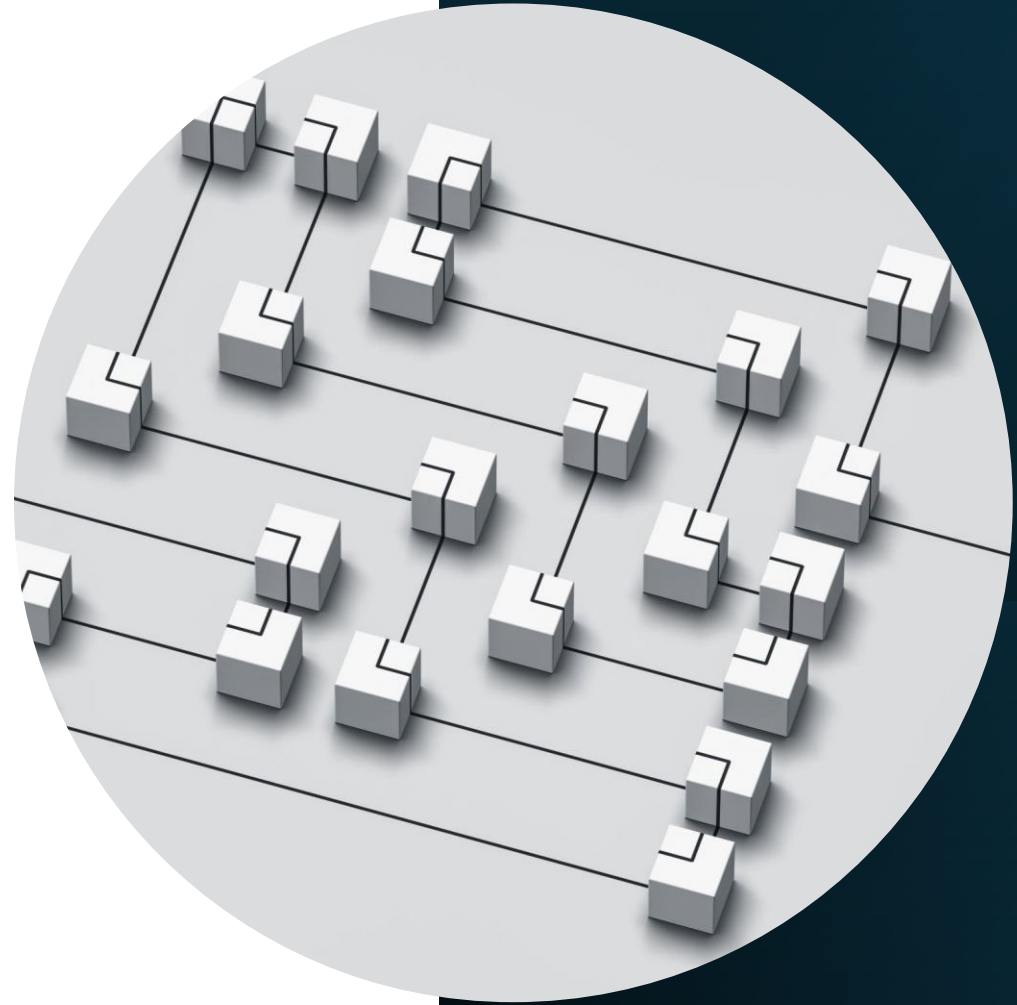


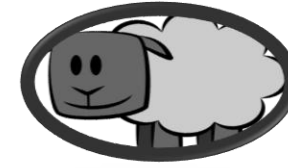
Basics of Building a Data Model in Power BI

Jason Romans



Jason Romans

Senior BI Engineer
Builder of Models



The Dax Shepherd



Lives in Nashville, Tennessee, United States



Started as SQL Server DBA



Transitioned to the Microsoft BI Stack



Work on everything from SQL Server Integration Services, SQL Server Database, Analysis Services, and Power BI



Favorite Data Model

Our Journey



1. Intro
2. Power Query
3. Data Model
4. External Tools
5. Conclusion

Our Journey



1. Intro

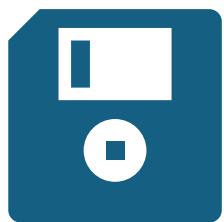
2. Power Query

3. Data Model

4. External Tools

5. Conclusion

Daunting Task



How do I bring in the data



What should the model
look like



Visualize Data

What you
picture



Reality



Encourage not discourage

WB Studio Tour

MODEL MAKING

'Be it model railways or dollhouses, models have played a part in all of our lives.

We can all relate to miniatures.'

– Gary Tomkins, Art Director, Models



Leavesden's Model Making Department constructed many of the wizarding world's most iconic locales as *picture models*, or perfect miniatures of the on-camera sets. Their full-colour, hand-painted models were crafted with the tiniest details, including foliage, stained glass and working lights. Along with Hogwarts castle, the talented crew also built miniatures of Hogsmeade village, the Owlery and the Weasley Burrow, a model that took 14 weeks to build and just six minutes to burn for *Harry Potter and the Half-Blood Prince*.



Instructions

- Unfortunately, no step-by-step instructions
- Lego blocks
- Guidelines

Our Journey



1. Intro

2. Power Query

3. Data Model

4. External Tools

5. Conclusion



Get Data

Where is the data



EXCEL SPREADSHEET

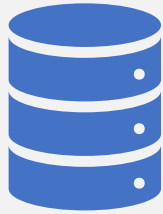


SQL SERVER

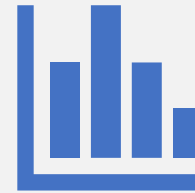


CSV

What form is the data in



SOURCE DATA THAT IS IN
SEPARATE TABLES

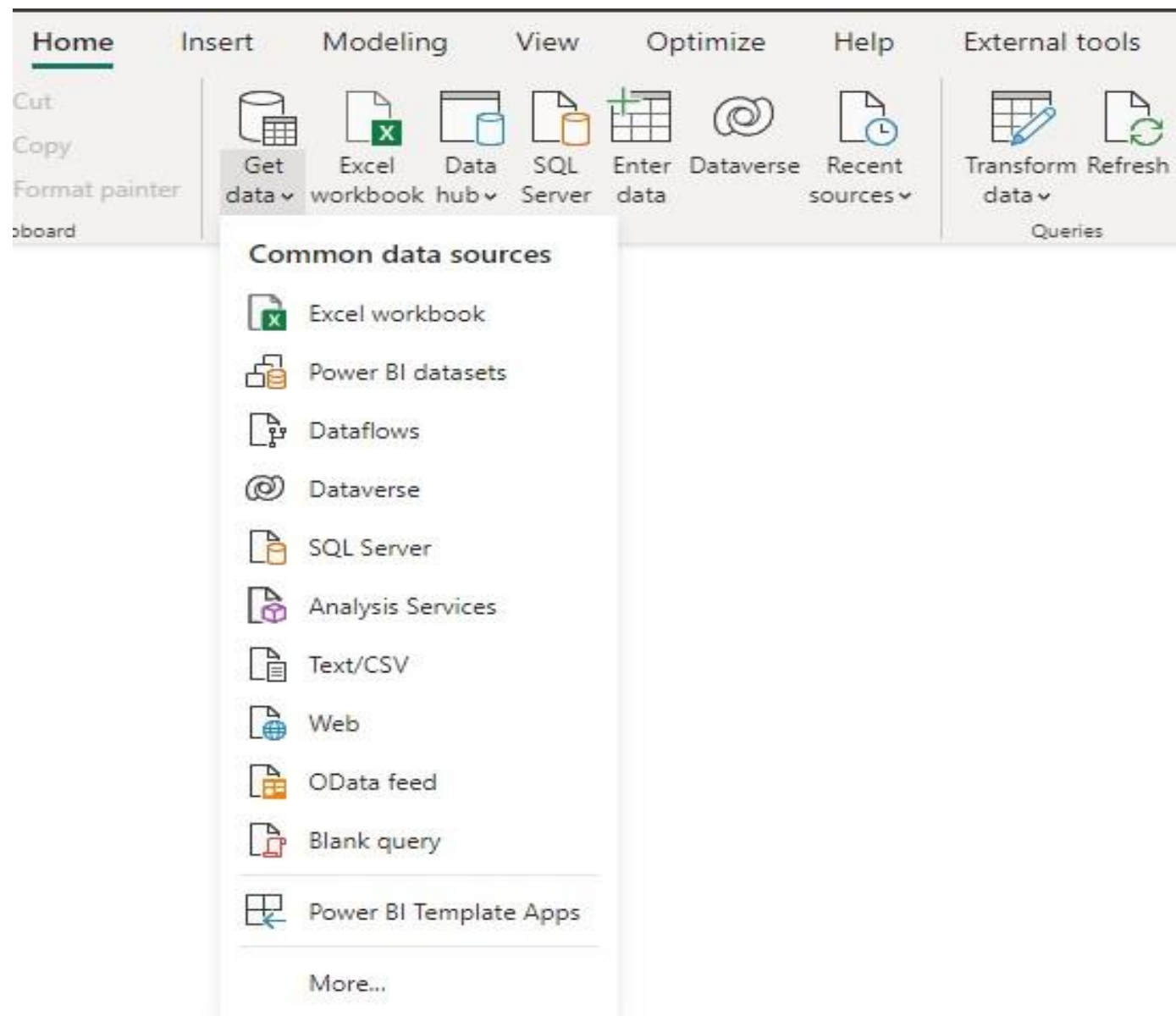


REPORT FORM WITH ALL
THE DATA COMBINED

Toy Model Builder

- Start - picture of finished product
- Proceed through instructions step by step
- Unlike toy model – changes can be made – no glue

Get Data



Import or DirectQuery

×

SQL Server database

Server ⓘ
Demo

Database (optional)
Contoso10K

Data Connectivity mode ⓘ
☒ Import
☐ DirectQuery

▸ Advanced options

OKCancel

Select Tables or Views

Navigator

Display Options ▾

demo: Contoso10K [20]

☐ Currency Exchange

☒ Customer

☒ Date

☐ DIM_Customer

☐ DIM_Date

☐ DIM_Product

☐ DIM_Store

☐ FACT_Sales

☒ Product

☒ Sales

☒ Store

☐ Data.CurrencyExchange

☐ Data.Customer

☐ Data.Date

☐ Data.GeoLocations

☐ Data.OrderRows

☐ Data.Orders

☐ Data.Product

☐ Data.Store

☐ fn IsHolidav

Sales

Order Number	Line Number	Order Date	Delivery Date	CustomerKey	StoreKey	ProductKey
36600	0	1/1/2018	1/1/2018	1620763	540	
36700	0	1/2/2018	1/2/2018	1320351	480	
36700	1	1/2/2018	1/2/2018	1320351	480	
36700	2	1/2/2018	1/2/2018	1320351	480	
36700	3	1/2/2018	1/2/2018	1320351	480	
36700	4	1/2/2018	1/2/2018	1320351	480	
36701	0	1/2/2018	1/2/2018	2077250	585	
36800	0	1/3/2018	1/10/2018	1743422	999999	
36800	1	1/3/2018	1/10/2018	1743422	999999	
36800	2	1/3/2018	1/10/2018	1743422	999999	
36800	3	1/3/2018	1/10/2018	1743422	999999	
36801	0	1/3/2018	1/3/2018	1013948	390	
36801	1	1/3/2018	1/3/2018	1013948	390	
36801	2	1/3/2018	1/3/2018	1013948	390	
36801	3	1/3/2018	1/3/2018	1013948	390	
36900	0	1/4/2018	1/4/2018	544050	220	
36901	0	1/4/2018	1/14/2018	485447	999999	
36902	0	1/4/2018	1/17/2018	1075463	999999	
37000	0	1/5/2018	1/5/2018	1319917	550	
37000	1	1/5/2018	1/5/2018	1319917	550	
37000	2	1/5/2018	1/5/2018	1319917	550	
37000	3	1/5/2018	1/5/2018	1319917	550	
37100	0	1/6/2018	1/6/2018	1107123	380	
37100	1	1/6/2018	1/6/2018	1107123	380	

Select Related Tables

Load

Transform Data

Cancel

Load or Transform Data

Navigator

Display Options ▾

demo: Contoso10K [20]

☐ Currency Exchange

☒ Customer

☒ Date

☐ DIM_Customer

☐ DIM_Date

☐ DIM_Product

☐ DIM_Store

☐ FACT_Sales

☒ Product

☒ Sales

☒ Store

☐ Data.CurrencyExchange

☐ Data.Customer

☐ Data.Date

☐ Data.GeoLocations

☐ Data.OrderRows

☐ Data.Orders

☐ Data.Product

☐ Data.Store

☐ fn IsHoliday

Sales

Order Number	Line Number	Order Date	Delivery Date	CustomerKey	StoreKey	ProductKey
36600	0	1/1/2018	1/1/2018	1620763	540	
36700	0	1/2/2018	1/2/2018	1320351	480	
36700	1	1/2/2018	1/2/2018	1320351	480	
36700	2	1/2/2018	1/2/2018	1320351	480	
36700	3	1/2/2018	1/2/2018	1320351	480	
36700	4	1/2/2018	1/2/2018	1320351	480	
36701	0	1/2/2018	1/2/2018	2077250	585	
36800	0	1/3/2018	1/10/2018	1743422	999999	
36800	1	1/3/2018	1/10/2018	1743422	999999	
36800	2	1/3/2018	1/10/2018	1743422	999999	
36800	3	1/3/2018	1/10/2018	1743422	999999	
36801	0	1/3/2018	1/3/2018	1013948	390	
36801	1	1/3/2018	1/3/2018	1013948	390	
36801	2	1/3/2018	1/3/2018	1013948	390	
36801	3	1/3/2018	1/3/2018	1013948	390	
36900	0	1/4/2018	1/4/2018	544050	220	
36901	0	1/4/2018	1/14/2018	485447	999999	
36902	0	1/4/2018	1/17/2018	1075463	999999	
37000	0	1/5/2018	1/5/2018	1319917	550	
37000	1	1/5/2018	1/5/2018	1319917	550	
37000	2	1/5/2018	1/5/2018	1319917	550	
37000	3	1/5/2018	1/5/2018	1319917	550	
37100	0	1/6/2018	1/6/2018	1107123	380	
37100	1	1/6/2018	1/6/2018	1107123	380	

Select Related Tables

Load

Transform Data

Cancel

Power Query Editor

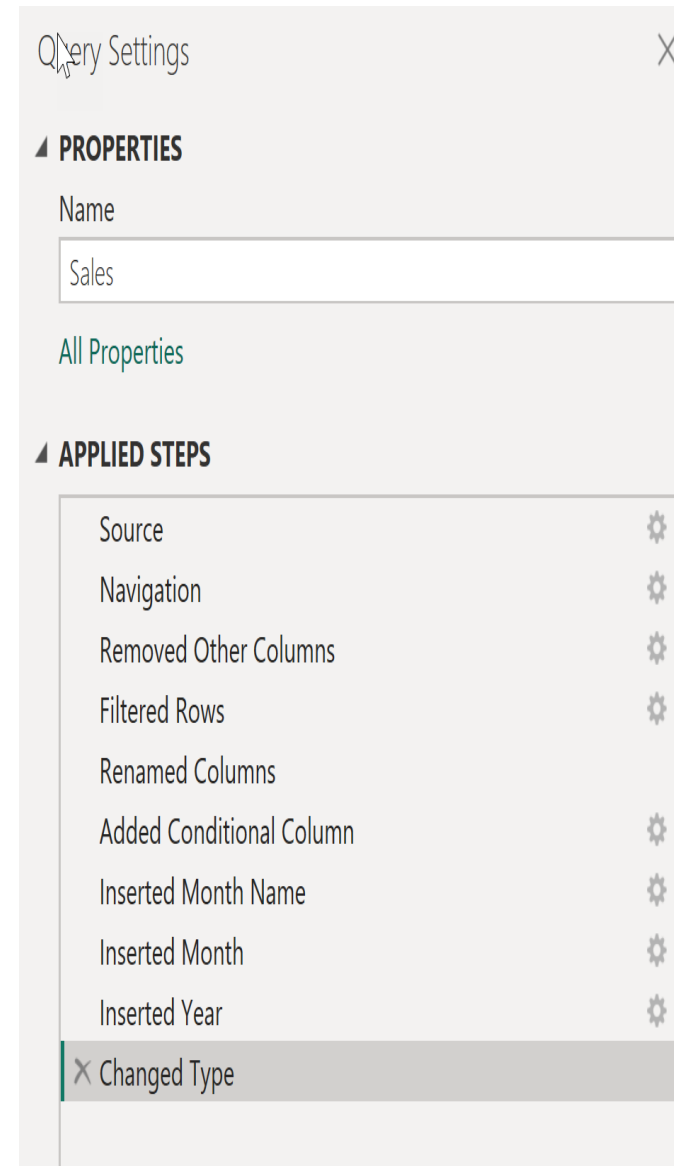
The screenshot displays the Microsoft Power Query Editor window. The title bar indicates the file is 'Untitled - Power Query Editor'. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing options like Split Column, Group By, and Replace Values. The left sidebar shows a list of queries: Store, Product, Sales (selected), and Customer. The main area displays a data table with the following columns: Order Number, Line Number, Order Date, Delivery Date, and CustomerKey. The table contains 25 rows of data. The right-hand pane shows the 'Query Settings' for the 'Sales' query, including the 'PROPERTIES' section with the name 'Sales' and the 'APPLIED STEPS' section with a single step named 'Source'.

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	36600	0	1/1/2018	1/1/2018	
2	36700	0	1/2/2018	1/2/2018	
3	36700	1	1/2/2018	1/2/2018	
4	36700	2	1/2/2018	1/2/2018	
5	36700	3	1/2/2018	1/2/2018	
6	36700	4	1/2/2018	1/2/2018	
7	36701	0	1/2/2018	1/2/2018	
8	36800	0	1/3/2018	1/10/2018	
9	36800	1	1/3/2018	1/10/2018	
10	36800	2	1/3/2018	1/10/2018	
11	36800	3	1/3/2018	1/10/2018	
12	36801	0	1/3/2018	1/3/2018	
13	36801	1	1/3/2018	1/3/2018	
14	36801	2	1/3/2018	1/3/2018	
15	36801	3	1/3/2018	1/3/2018	
16	36900	0	1/4/2018	1/4/2018	
17	36901	0	1/4/2018	1/14/2018	
18	36902	0	1/4/2018	1/17/2018	
19	37000	0	1/5/2018	1/5/2018	
20	37000	1	1/5/2018	1/5/2018	
21	37000	2	1/5/2018	1/5/2018	
22	37000	3	1/5/2018	1/5/2018	
23	37100	0	1/6/2018	1/6/2018	
24	37100	1	1/6/2018	1/6/2018	
25					

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2:04 PM

Power Query Steps like a Recipe



Promote First Row to Headers (CSV)

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By

Use First Row as Headers

Transpose

Reverse Rows

Count Rows

Use First Row as Headers

Use Headers as First Row

Data Type: Text

Detect Data Type

Rename

Replace Values

Fill

Pivot Column

Unpivot Columns

Move

Convert to List

Any Column

Promote the first row of this table into column headers.

Sales

	Column1	Column2
1	Order Number	Line Number
2	36600	0
3	36700	0
4	36700	1
5	36700	2

Promote First Row to Headers (CSV)

The screenshot displays the Power Query Editor interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing options like 'Use First Row as Headers' and 'Replace Values'. The main area shows a table with 13 columns and 999+ rows. The first row is highlighted in green, indicating it has been promoted to headers. The formula bar at the top shows the query definition: `= Csv.Document(File.Contents("C:\ContosoDataFiles\Sales.csv"),[Delimiter=",",`. The right sidebar shows the 'Query Settings' pane with the 'Name' field set to 'Sales' and the 'Applied Steps' list containing 'Source'.

Order Number	Line Number	Order Date	Delivery Date
36600	0	2018-01-01	2018-01-01
36700	0	2018-01-02	2018-01-02
36700	1	2018-01-02	2018-01-02
36700	2	2018-01-02	2018-01-02
36700	3	2018-01-02	2018-01-02
36700	4	2018-01-02	2018-01-02
36701	0	2018-01-02	2018-01-02
36800	0	2018-01-03	2018-01-10
36800	1	2018-01-03	2018-01-10
36800	2	2018-01-03	2018-01-10
36800	3	2018-01-03	2018-01-10
36801	0	2018-01-03	2018-01-03
36801	1	2018-01-03	2018-01-03
36801	2	2018-01-03	2018-01-03
36801	3	2018-01-03	2018-01-03
36900	0	2018-01-04	2018-01-04
36901	0	2018-01-04	2018-01-14

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 1:27 PM

Promote First Row to Headers (CSV)

The screenshot shows the Power Query Editor interface. The formula bar at the top displays the M formula: `= Table.PromoteHeaders(Source, [PromoteAllScalars=true])`. Below the formula bar, a table of data is shown. The first row of the table is highlighted with a red border, indicating it is the row being promoted to headers. The table has four columns: Order Number, Line Number, Order Date, and Delivery Date. The data rows follow, starting with Order Number 36600 and Line Number 0.

	Order Number	Line Number	Order Date	Delivery Date
1	36600	0	2018-01-01	2018-01-01
2	36700	0	2018-01-02	2018-01-02
3	36700	1	2018-01-02	2018-01-02
4	36700	2	2018-01-02	2018-01-02
5	36700	3	2018-01-02	2018-01-02
6	36700	4	2018-01-02	2018-01-02
7	36701	0	2018-01-02	2018-01-02
8	36800	0	2018-01-03	2018-01-10
9	36800	1	2018-01-03	2018-01-10
10	36800	2	2018-01-03	2018-01-10
11	36800	3	2018-01-03	2018-01-10
12	36801	0	2018-01-03	2018-01-03
13	36801	1	2018-01-03	2018-01-03
14	36801	2	2018-01-03	2018-01-03
15	36801	3	2018-01-03	2018-01-03
16	36900	0	2018-01-04	2018-01-04
17	36901	0	2018-01-04	2018-01-14
18	36902	0	2018-01-04	2018-01-17

At the bottom of the window, it says "13 COLUMNS, 999+ ROWS" and "Column profiling based on top 1000 rows". The status bar also indicates "PREVIEW DOWNLOADED AT 1:28 PM".

Remove Other Columns

The screenshot displays the Microsoft Power Query Editor window titled "Untitled - Power Query Editor". The ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The "Remove Columns" menu is open, showing options: "Remove Columns", "Remove Rows", "Remove Other Columns", and "Remove Columns". The main data area shows a table with columns: Order Number, Line Number, Order Date, Delivery Date, and CustomerKey. The table contains 25 rows of data. The right sidebar shows the "Query Settings" pane with "Properties" and "Applied Steps" sections. The "Applied Steps" section shows a list of steps: "Source" and "Navigation".

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	36600	0	1/1/2018	1/1/2018	1620763
2	36700	0	1/2/2018	1/2/2018	1320351
3	36700	1	1/2/2018	1/2/2018	1320351
4	36700	2	1/2/2018	1/2/2018	1320351
5	36700	3	1/2/2018	1/2/2018	1320351
6	36700	4	1/2/2018	1/2/2018	1320351
7	36701	0	1/2/2018	1/2/2018	2077250
8	36800	0	1/3/2018	1/10/2018	1743422
9	36800	1	1/3/2018	1/10/2018	1743422
10	36800	2	1/3/2018	1/10/2018	1743422
11	36800	3	1/3/2018	1/10/2018	1743422
12	36801	0	1/3/2018	1/3/2018	1013948
13	36801	1	1/3/2018	1/3/2018	1013948
14	36801	2	1/3/2018	1/3/2018	1013948
15	36801	3	1/3/2018	1/3/2018	1013948
16	36900	0	1/4/2018	1/4/2018	544050
17	36901	0	1/4/2018	1/14/2018	485447
18	36902	0	1/4/2018	1/17/2018	1075463
19	37000	0	1/5/2018	1/5/2018	1319917
20	37000	1	1/5/2018	1/5/2018	1319917
21	37000	2	1/5/2018	1/5/2018	1319917
22	37000	3	1/5/2018	1/5/2018	1319917
23	37100	0	1/6/2018	1/6/2018	1107123
24	37100	1	1/6/2018	1/6/2018	1107123
25					

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON TUESDAY

Filter Rows

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Data Sources Manage Parameters Parameters Refresh Preview Advanced Editor Query Choose Columns Remove Columns Manage Columns Keep Rows Remove Rows Reduce Rows Sort Split Column Group By Data Type: Date Use First Row as Headers Replace Values Transform Combine Text Analytics Vision Azure Machine Learning All Insights

Queries [1] Sales

Order Number Line Number Order Date Delivery Date CustomerKey

1 36600 0 1/1/2018 1/1/2018 1620763

2 36700 0 1/2/2018 1/2/2018 1320351

3 36700 1 1/2/2018 1/2/2018 1320351

4 36700 2 1/2/2018 1/2/2018 1320351

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16

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20 37000 1 1/5/2018 1/5/2018 1319917

21 37000 2 1/5/2018 1/5/2018 1319917

22 37000 3 1/5/2018 1/5/2018 1319917

23 37100 0 1/6/2018 1/6/2018 1107123

24 37100 1 1/6/2018 1/6/2018 1107123

25

Filter Rows

Apply one or more filter conditions to the rows in this table.

☒ Basic ☐ Advanced

Keep rows where 'Order Date'

is after 6/13/2018

☒ And ☐ Or

Enter or select a value

OK Cancel

Query Settings

PROPERTIES

Name Sales

APPLIED STEPS

Source

X Navigation

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON TUESDAY

Set Data Types

- Correct types matter
 - Same type for relationships
- Performance
- Calculations

ProductKey as a String

Name	Cardinality	Total Size ↓	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table
Product	2,517	15,142,472	27,520	15,043,768	71,184	Many	-	-	0	0	
Product Name	2,517	1,128,480	4,160	1,104,176	20,144	HASH	String	-	-	-	7.45%
Product Code	2,517	1,127,544	4,160	1,103,240	20,144	HASH	String	-	-	-	7.45%
ProductKey	2,517	1,127,416	4,160	1,103,112	20,144	HASH	String	-	-	-	7.45%
Unit Cost	480	1,076,408	3,008	1,069,544	3,856	HASH	String	-	-	-	7.11%
Unit Price	426	1,075,520	3,008	1,069,088	3,424	HASH	String	-	-	-	7.10%
Weight	297	1,073,176	2,736	1,068,056	2,384	HASH	String	-	-	-	7.09%
Subcategory	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Subcategory Code	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Color	16	1,067,144	1,192	1,065,808	144	HASH	String	-	-	-	7.05%
Brand	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Manufacturer	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Category	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Category Code	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Weight Unit Measure	4	1,065,896	136	1,065,712	48	HASH	String	-	-	-	7.04%

ProductKey as a Whole Number

Name	Cardinality	Total Size ↓	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table
Product	2,517	14,115,268	27,520	14,016,564	71,184	Many	-	-	0	0	
Product Name	2,517	1,128,480	4,160	1,104,176	20,144	HASH	String	-	-	-	7.99%
Product Code	2,517	1,127,544	4,160	1,103,240	20,144	HASH	String	-	-	-	7.99%
Unit Cost	480	1,076,408	3,008	1,069,544	3,856	HASH	String	-	-	-	7.63%
Unit Price	426	1,075,520	3,008	1,069,088	3,424	HASH	String	-	-	-	7.62%
Weight	297	1,073,176	2,736	1,068,056	2,384	HASH	String	-	-	-	7.60%
Subcategory	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.56%
Subcategory Code	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.56%
Color	16	1,067,144	1,192	1,065,808	144	HASH	String	-	-	-	7.56%
Brand	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.56%
Manufacturer	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.56%
Category	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.55%
Category Code	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.55%
Weight Unit Measure	4	1,065,896	136	1,065,712	48	HASH	String	-	-	-	7.55%
ProductKey	2,517	100,212	4,160	75,908	20,144	HASH	Int64	-	-	-	0.71%

Cardinality

Name	Cardinality	Total Size ↓	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table
Product	2,517	15,142,472	27,520	15,043,768	71,184	Many	-	-	0	0	
Product Name	2,517	1,128,480	4,160	1,104,176	20,144	HASH	String	-	-	-	7.45%
Product Code	2,517	1,127,544	4,160	1,103,240	20,144	HASH	String	-	-	-	7.45%
ProductKey	2,517	1,127,416	4,160	1,103,112	20,144	HASH	String	-	-	-	7.45%
Unit Cost	480	1,076,408	3,008	1,069,544	3,856	HASH	String	-	-	-	7.11%
Unit Price	426	1,075,520	3,008	1,069,088	3,424	HASH	String	-	-	-	7.10%
Weight	297	1,073,176	2,736	1,068,056	2,384	HASH	String	-	-	-	7.09%
Subcategory	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Subcategory Code	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Color	16	1,067,144	1,192	1,065,808	144	HASH	String	-	-	-	7.05%
Brand	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Manufacturer	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Category	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Category Code	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Weight Unit Measure	4	1,065,896	136	1,065,712	48	HASH	String	-	-	-	7.04%

Cardinality – Unique Values

- The less unique values the better
- Do you really need Date and Time
- Does the time need to include seconds
 - What about morning, afternoon, night

Custom Columns

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Store Size

	Column Name	Operator	Value ⓘ			Output ⓘ		
If	Square Meters ▾	equals ▾	ABC 123 ▾	null	Then	ABC 123 ▾	N/A	...
Else If	Square Meters ▾	is less than or equ... ▾	ABC 123 ▾	500	Then	ABC 123 ▾	Small	...
Else If	Square Meters ▾	is less than or equ... ▾	ABC 123 ▾	1000	Then	ABC 123 ▾	Medium	

Add Clause

Else ⓘ

ABC 123 ▾

Large

OK

Cancel

Add Column from Example

Add Column From Examples

Enter sample values to create a new column (Ctrl+Enter to apply).

	A ^B _C Zip Code	A ^B _C Country Code	A ^B _C Country	A ^B _C Continent	A ^B _C Birthday
1	13901	US	United States	North America	5/14/1982 0:00
2	99362	US	United States	North America	6/25/1947 0:00
3	47274	US	United States	North America	8/29/1942 0:00
4	48607	US	United States	North America	8/16/1946 0:00
5	36330	US	United States	North America	12/19/1992 0:00
6	76102	US	United States	North America	3/3/1946 0:00
7	91762	US	United States	North America	10/3/2001 0:00
8	20036	US	United States	North America	3/15/1976 0:00
9	07501	US	United States	North America	4/9/1979 0:00
10	52501	US	United States	North America	9/3/1947 0:00
11	78040	US	United States	North America	10/3/1995 0:00
12	71201	US	United States	North America	2/13/1997 0:00
13	53916	US	United States	North America	4/28/1964 0:00
14	85283	US	United States	North America	1/2/1953 0:00
15	63011	US	United States	North America	1/13/1956 0:00
16	97205	US	United States	North America	12/21/1971 0:00
17	39211	US	United States	North America	11/24/1998 0:00
18	85364	US	United States	North America	10/12/1946 0:00
19	03276	US	United States	North America	5/29/1938 0:00
20	95204	US	United States	North America	2/13/1982 0:00
21	85724	US	United States	North America	11/19/1943 0:00
22	97205	US	United States	North America	10/19/1975 0:00

OK
Cancel

Column1
▼

- 5/14/1982 0:00 (Birthday)
- 14 (Day from Birthday)
- 5 (Day of Week from Birthday)
- Friday (Day of Week Name from Birthday)
- 134 (Day of Year from Birthday)
- 31 (Days in Month from Birthday)
- 1982-05-14T23:59:59.9999999 (End of Day from Birthday)
- 1982-05-31T23:59:59.9999999 (End of Month from Birthday)
- 1982-06-30T23:59:59.9999999 (End of Quarter from Birthday)
- 1982-05-15T23:59:59.9999999 (End of Week from Birthday)
- 1982-12-31T23:59:59.9999999 (End of Year from Birthday)
- 5 (Month from Birthday)
- May (Month Name from Birthday)
- 2 (Quarter of Year from Birthday)
- 5/1/1982 12:00:00 AM (Start of Month from Birthday)
- 4/1/1982 12:00:00 AM (Start of Quarter from Birthday)
- 5/9/1982 12:00:00 AM (Start of Week from Birthday)
- 1/1/1982 12:00:00 AM (Start of Year from Birthday)
- 3 (Week of Month from Birthday)
- 20 (Week of Year from Birthday)

Query Settings

PROPERTIES

Name
Customer

All Properties

Add Column from Example

Add Column From Examples

Enter sample values to create a new column (Ctrl+Enter to apply).

Transform: `Date.Year(DateTime.From([Birthday]))`

OK

Cancel

	☒ Zip Code	☐ Country Code	☐ Country	☐ Continent	☒ Birthday	Year
1	13901	US	United States	North America	5/14/1982 0:00	1982
2	99362	US	United States	North America	6/25/1947 0:00	1947
3	47274	US	United States	North America	8/29/1942 0:00	1942
4	48607	US	United States	North America	8/16/1946 0:00	1946
5	36330	US	United States	North America	12/19/1992 0:00	1992
6	76102	US	United States	North America	3/3/1946 0:00	1946
7	91762	US	United States	North America	10/3/2001 0:00	2001
8	20036	US	United States	North America	3/15/1976 0:00	1976
9	07501	US	United States	North America	4/9/1979 0:00	1979
10	52501	US	United States	North America	9/3/1947 0:00	1947
11	78040	US	United States	North America	10/3/1995 0:00	1995
12	71201	US	United States	North America	2/13/1997 0:00	1997
13	53916	US	United States	North America	4/28/1964 0:00	1964
14	85283	US	United States	North America	1/2/1953 0:00	1953
15	63011	US	United States	North America	1/13/1956 0:00	1956
16	97205	US	United States	North America	12/21/1971 0:00	1971
17	39211	US	United States	North America	11/24/1998 0:00	1998
18	85364	US	United States	North America	10/12/1946 0:00	1946
19	03276	US	United States	North America	5/29/1938 0:00	1938
20	95204	US	United States	North America	2/13/1982 0:00	1982
21	85724	US	United States	North America	11/19/1943 0:00	1943
22	97205	US	United States	North America	10/19/1975 0:00	1975
23	48066	US	United States	North America	8/13/1995 0:00	1995
24	38049	US	United States	North America	11/13/1939 0:00	1939
25	13202	US	United States	North America	11/24/1961 0:00	1961
26	49546	US	United States	North America	8/24/1940 0:00	1940
27	60014	US	United States	North America	6/17/1985 0:00	1985
28	32501	US	United States	North America	4/16/1951 0:00	1951
29	99336	US	United States	North America	6/22/1999 0:00	1999

Roche's
Maxim

***Data should be
transformed as far
upstream as possible,
and as far downstream as
necessary.***

--Matthew Roche

<https://ssbipolar.com/2021/05/31/roches-maxim/>

Demo

Query Folding

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Replace Values Data Type: Any Use First Row as Headers Combine Text Analytics Vision Azure Machine Learning AI Insights

Queries [1] Sales

= Table.SelectRows("#Removed Other Columns", each [Order Date] > #date(2018, 6, 13))

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	53000	0	6/14/2018	6/14/2018	1365902
2	53000	1	6/14/2018	6/14/2018	1365902
3	53200	0	6/16/2018	6/16/2018	1656264
4	53200	1	6/16/2018	6/16/2018	1656264
5	53200	2	6/16/2018	6/16/2018	1656264
6	53200	3	6/16/2018	6/16/2018	1656264
7	53600	0	6/20/2018	7/2/2018	1799984
8	53700	0	6/21/2018	6/21/2018	1373034
9	53700	1	6/21/2018	6/21/2018	1373034
10	53700	2	6/21/2018	6/21/2018	1373034
11	53900	0	6/23/2018	6/23/2018	326607
12	53900	1	6/23/2018	6/23/2018	326607
13	53900	2	6/23/2018	6/23/2018	326607
14	54300	0	6/27/2018	6/27/2018	687715
15	54400	0	6/28/2018	6/28/2018	1734949
16	54400	1	6/28/2018	6/28/2018	1734949
17	54600	0	6/30/2018	6/30/2018	1043780
18	54600	1	6/30/2018	6/30/2018	1043780
19	55000	0	7/4/2018	7/4/2018	98175
20	55000	1	7/4/2018	7/4/2018	98175
21	55000	2	7/4/2018	7/4/2018	98175
22	55000	3	7/4/2018	7/4/2018	98175
23	55000	4	7/4/2018	7/4/2018	98175
24	55000	5	7/4/2018	7/4/2018	98175
25	55000	6	7/4/2018	7/4/2018	98175

5 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2:06 PM

Query Settings

PROPERTIES

Name: Sales

APPLIED STEPS

Source

Navigation

Removed Other Columns

Filter

Edit Settings

Rename

Delete

Delete Until End

Insert Step After

Move before

Move after

Extract Previous

View Native Query

Diagnose

Properties...

Native Query

Native Query

```
select [_].[Order Number],  
       [_].[Line Number],  
       [_].[Order Date],  
       [_].[Delivery Date],  
       [_].[CustomerKey]  
from  
(  
    select [Order Number],  
           [Line Number],  
           [Order Date],  
           [Delivery Date],  
           [CustomerKey]  
    from [dbo].[Sales] as [$Table]  
) as [_]  
where [_].[Order Date] > convert(date, '2018-06-13')
```

OK

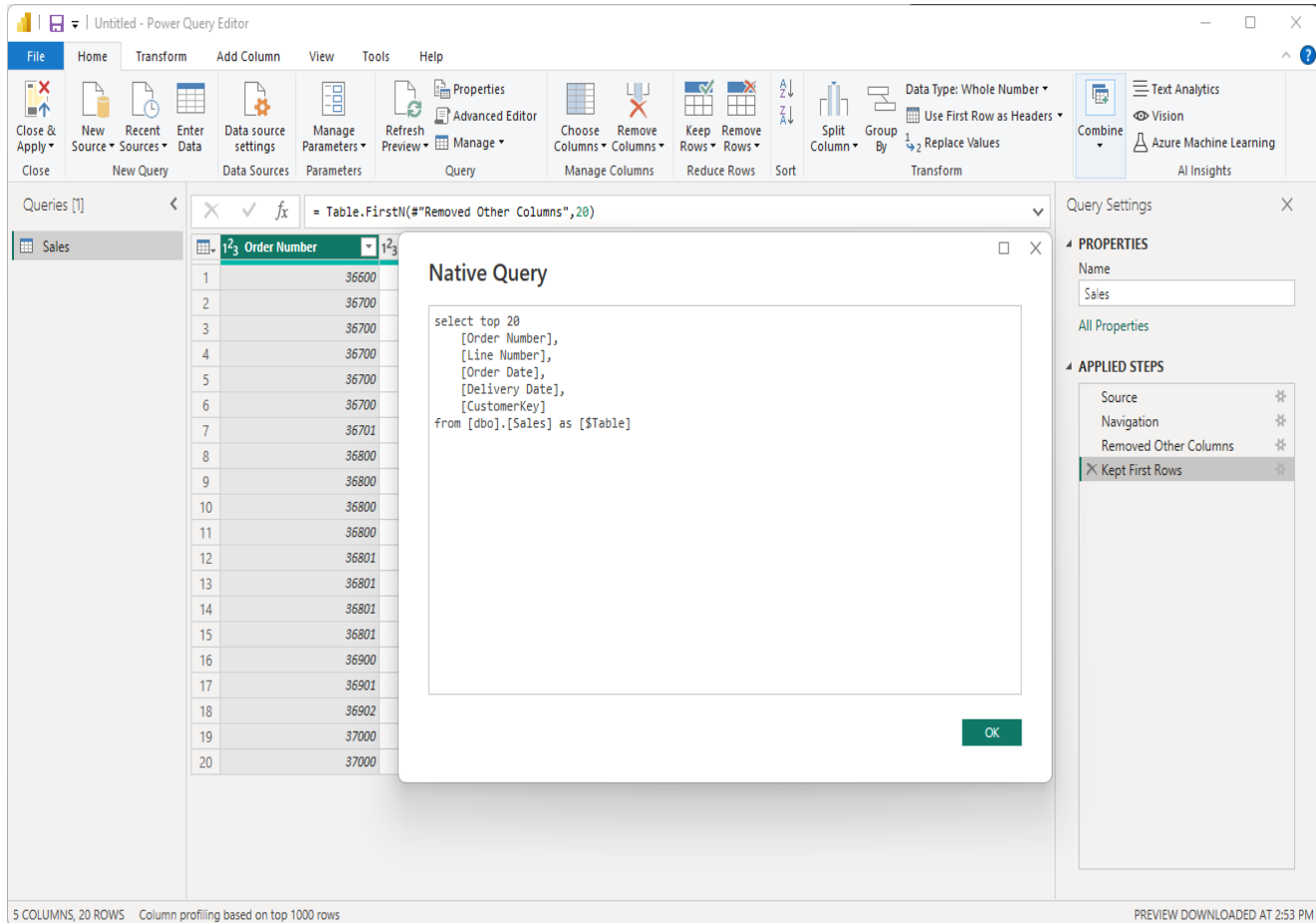
Breaking (Bad) Query Folding

The screenshot shows the Power Query Editor interface. The formula bar at the top displays the query formula: `= Table.LastN(#"Removed Other Columns", 20)`. The data table below shows 20 rows of data with columns: Order Number, Line Number, Order Date, Delivery Date, and CustomerKey. The 'APPLIED STEPS' pane on the right shows the step 'Removed Other Columns' selected, with a context menu open showing options like 'View Native Query'.

Order Number	Line Number	Order Date	Delivery Date	CustomerKey
115302	1	2/27/2020	2/27/2020	924770
115400	0	2/28/2020	2/28/2020	353571
115400	1	2/28/2020	2/28/2020	353571
115401	0	2/28/2020	3/8/2020	1358107
115500	0	2/29/2020	2/29/2020	345884
115501	0	2/29/2020	3/2/2020	1485625
115501	1	2/29/2020	3/2/2020	1485625
115501	2	2/29/2020	3/2/2020	1485625
115501	3	2/29/2020	3/2/2020	1485625
115501	4	2/29/2020	3/2/2020	1485625
115501	5	2/29/2020	3/2/2020	1485625
115501	6	2/29/2020	3/2/2020	1485625
115502	0	2/29/2020	2/29/2020	982512
115502	1	2/29/2020	2/29/2020	982512
115502	2	2/29/2020	2/29/2020	982512
115502	3	2/29/2020	2/29/2020	982512
115502	4	2/29/2020	2/29/2020	982512
115502	5	2/29/2020	2/29/2020	982512
115502	6	2/29/2020	2/29/2020	982512
115800	0	3/3/2020	3/3/2020	237000

Keep bottom 20 rows breaks query folding
SQL Server does not have a bottom clause

Query Folding



Keep Top 20 Rows does not
SQL Server has a top N clause

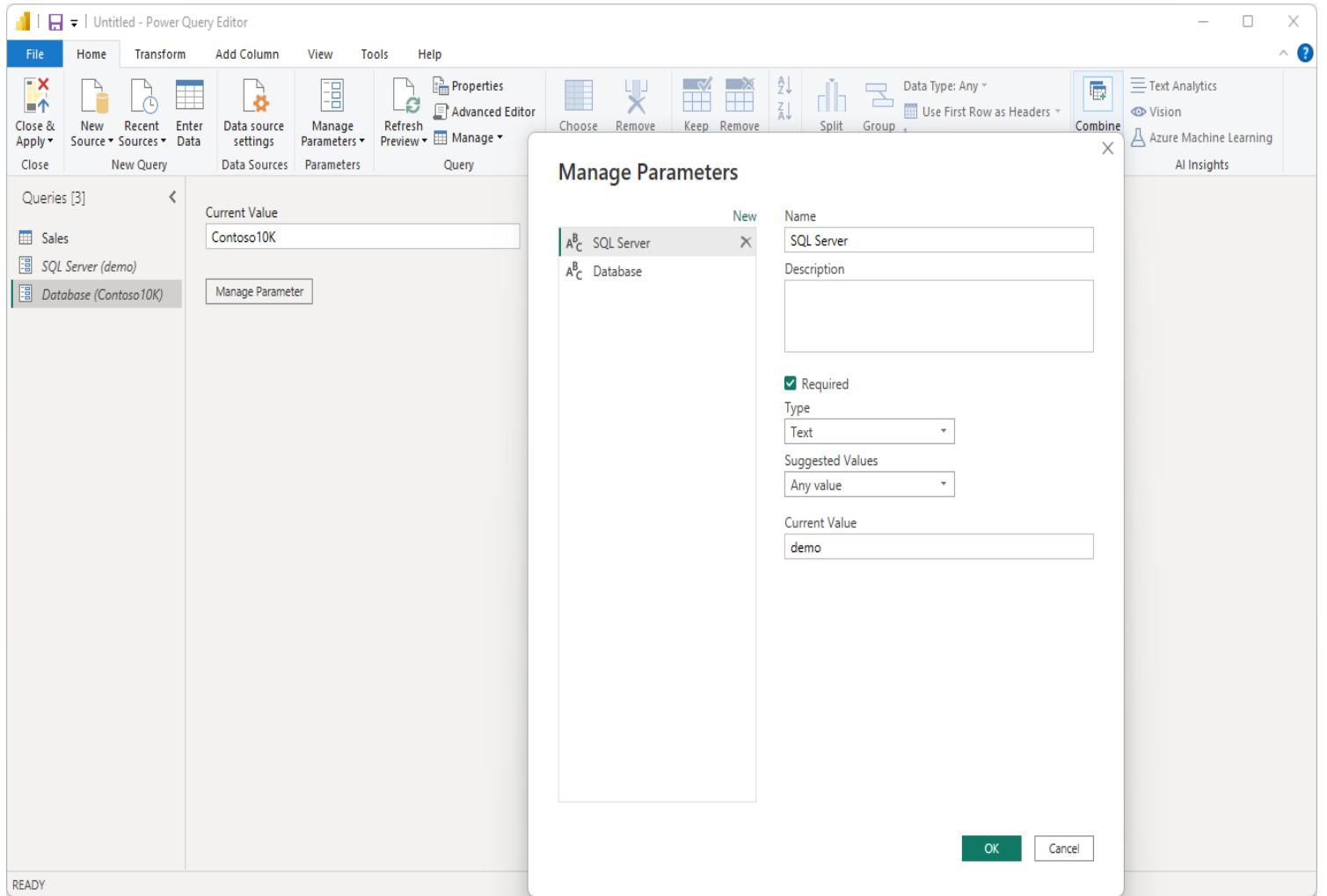
Query Folding

- This is the query sent when the model is refreshed
- This is not the query sent to the tabular engine when you interact with the visuals
- Put off breaking the query folding till the later steps

Power Query in Power BI Desktop

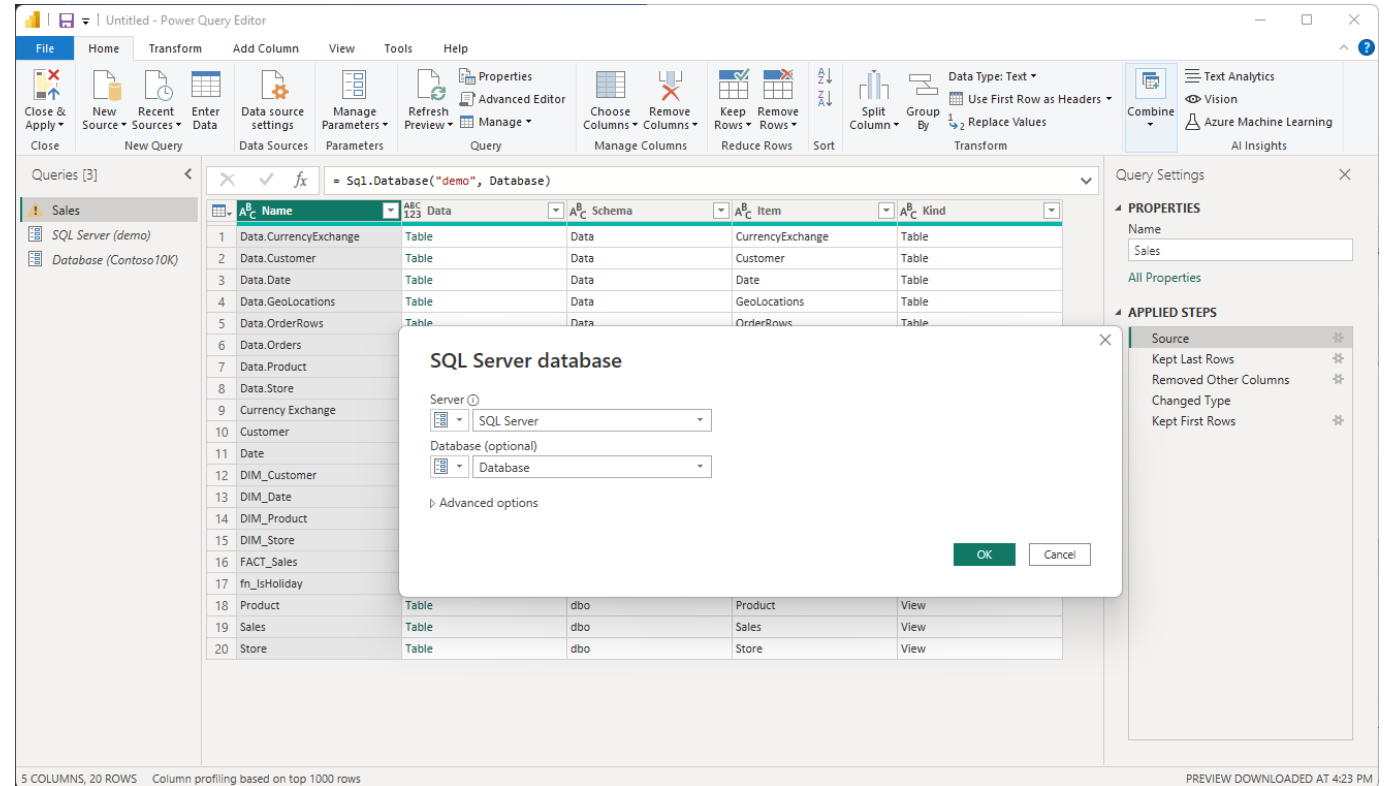
- New Experience with Icons like Power Query Online
- Projected to be released this year

Parameters



Creating parameters

Using Parameters



Using a parameter for the Server and Database

Goals of Power Query Steps

- Get smallest dataset possible early in query steps
 - Remove Columns
 - Filter Rows
- Unique Values
- Preserve Query Folding as long as possible

Our Journey



1. Intro

2. Power Query

3. Data Model

4. External Tools

5. Conclusion

Data Model

Date Tables

Turn off Auto Date Tables

Options

GLOBAL

Data Load

Power Query Editor

DirectQuery

R scripting

Python scripting

Security

Privacy

Regional Settings

Updates

Usage Data

Diagnostics

Preview features

Auto recovery

Report settings

CURRENT FILE

Data Load

Regional Settings

Privacy

Auto recovery

Published dataset settings

Query reduction

Report settings

Background Data

☐ Always allow data previews to download in the background

☒ Allow data previews to download in the background according to each file's setting

☐ Never allow data previews to download in the background

Parallel loading of tables

When you load data into Power BI (via import or DirectQuery), each data table is backed by a Power Query query. These queries are evaluated simultaneously instead of one-by-one, which can speed up the process. In certain situations, you might want to adjust the default number of simultaneous query evaluations and memory used. [Learn more](#)

Maximum number of simultaneous evaluations

20

Maximum memory used per simultaneous evaluation (MB)

432

Time intelligence

☐ Auto date/time for new files

[Learn more](#)

Data Cache Management Options

Currently used: 655 KB

Clear Cache

Maximum allowed (MB):

4096

Restore Defaults

Q&A Cache Options

Currently used: 1.43 MB

Clear Cache

Maximum allowed (MB):

4096

OK

Cancel

Hidden Date Tables

- >  Customer
- >  Date
- ✓  *DateTableTemplate_a987aaf9-4324-4fff-9e6b-91eb917a6145*
 -  *Date*
 - >  Date Hierarchy
 - 1²₃ Day*
 - A^B_C Month*
 - 1²₃ MonthNo*
 - A^B_C Quarter*
 - 1²₃ QuarterNo*
 - 1²₃ Year*
- >  *LocalDateTable_329979ec-f948-448d-8d40-da32309c689b*
- >  *LocalDateTable_565d20a7-37da-41eb-acea-3d1681afac62*
- >  *LocalDateTable_6082b466-aa3a-416d-92f5-dee233105c41*
- >  *LocalDateTable_729c8b24-1385-4545-897f-e90fb9602a1d*
- >  *LocalDateTable_86220e95-4654-4888-819a-8afd2ddbe1da*
- >  *LocalDateTable_8ce283be-c847-412c-907b-87781a0d19e1*
- >  Product
- >  Sales
- >  Store

Date Table Alternatives



A spiral-bound desk calendar is shown, displaying a date table. The calendar is open to a page showing the days of the week (SUN, MON, TUE, WED, THU, FRI, SAT) and the dates (1 through 31). The calendar is placed on a white surface, and its reflection is visible below it.

SUN	MON	TUE	WED	THU	FRI	SAT
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Power Query (M): Date Table

Kristyna Ferris

- Date Table Built in Power Query
- <https://github.com/Anytsirk12/DataOnWheels/tree/main/SQL Bits%202024>



M Code

```
//----- M Pre-flight Checklist -----  
//Must start with a "let" statement and end with an "in" statement  
//Every step must end with a comma except the last step  
//When using Table. functions, be sure to reference the previous step name as the table.  
//Not whitespace sensitive, but it is case sensitive.  
//Always name your steps, makes troubleshooting and updating a million times easier!  
//To avoid extra steps, assign data types when creating a column.  
//Follow the DRY philosophy - Don't Repeat Yourself. Use variables and already created columns to avoid having to redo logic  
//If you want to remove columns, do so within power query since it will automatically fix the references in future steps  
  
let  
//Use variables in M to set the start date  
StartDate = #date(2021, 12, 1),  
//EndDate is dynamically the start of next year. You can hard code this like StartDate, but that's not recommended.  
EndDate = #date(Date.Year(DateTime.LocalNow())+1,1,1),  
Length = Duration.Days(EndDate - StartDate),  
Today = DateTime.LocalNow(),  
  
//Create list of dates using our variables. #duration increments by (days,hours,minutes,seconds)  
Source = List.Dates(StartDate, Length, #duration(1, 0, 0, 0)),  
  
//Convert this list to a table in order to add additional columns  
#"Converted to Table" = Table.FromList(Source, Splitter.SplitByNothing(), null, null, ExtraValues.Error),  
#"Renamed Date Column" = Table.RenameColumns("#Converted to Table",{{"Column1", "Date"}}),  
#"Changed Type to Date" = Table.TransformColumnTypes("#Renamed Date Column",{{"Date", type date}}),  
  
//Create a datekey for easy joins to fact tables  
#"Inserted DateKey" = Table.AddColumn("#Changed Type to Date", "DateKey", each DateTime.Date([Date]), type date),  
#"Changed DateKey to Int" = Table.TransformColumnTypes("#Inserted DateKey",{{"DateKey", Int64.Type}}),  
  
//Create columns for year & month  
#"Inserted Year" = Table.AddColumn("#Changed DateKey to Int", "Year", each Date.Year([Date]), Int64.Type),  
#"Inserted Month" = Table.AddColumn("#Inserted Year", "Month", each Date.Month([Date]), Int64.Type),  
#"Inserted Month Name" = Table.AddColumn("#Inserted Month", "Month Name", each Date.MonthName([Date]), type text),  
//MMM is very useful for visuals since it eliminates a lot of characters and saves space  
#"Inserted MMM" = Table.AddColumn("#Inserted Month Name", "MMM", each Text.Start([Month Name], 3), type text),  
#"Added Month Year" = Table.AddColumn("#Inserted MMM", "Month Year", each Text.Combine({[MMM], Text.From([Year], "en-US"))}, " "), type text),  
  
//Create columns for day & day of week  
#"Inserted Day" = Table.AddColumn("#Added Month Year", "Day", each Date.Day([Date]), Int64.Type),  
#"Inserted Day Name" = Table.AddColumn("#Inserted Day", "Day Name", each Date.DayOfWeekName([Date]), type text),  
//Sets Sunday as 1  
#"Inserted Day of Week" = Table.AddColumn("#Inserted Day Name", "Day of Week", each Date.DayOfWeek([Date], Day.Sunday)+1, Int64.Type),  
#"Inserted DDD" = Table.AddColumn("#Inserted Day of Week", "DDD", each Text.Start([Day Name], 3), type text),  
  
//Create MonthDay to use in visuals  
#"Inserted MonthDay" = Table.AddColumn("#Inserted DDD", "Month Day", each Text.Combine({[MMM],  
    if Text.Length(Text.From([Day])) = 1 then Text.Combine({"0",Text.From([Day]},"") else Text.From([Day]), " "), type text),  
//Need MonthDaySort to avoid visuals sorting alphabetically  
#"Inserted MonthDaySort" = Table.AddColumn("#Inserted MonthDay", "Month Day Sort", each Text.Combine({  
    if Text.Length(Text.From([Month]))=1 then Text.Combine({"0",Text.From([Month]},"") else Text.From([Month]),  
    if Text.Length(Text.From([Day])) = 1 then Text.Combine({"0",Text.From([Day]},"") else Text.From([Day])  
    }, ""),type text),
```

Date Table View

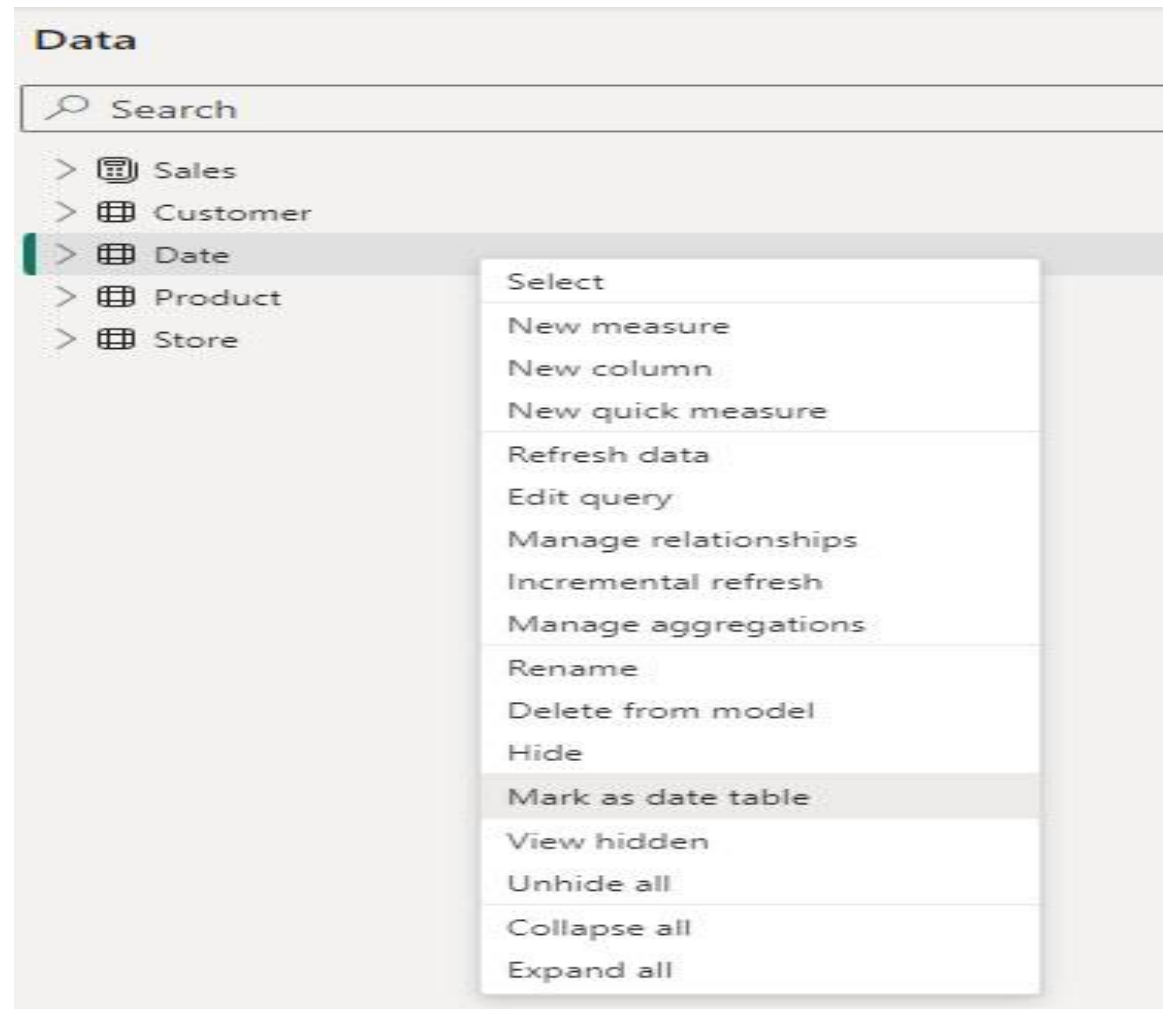
CalendarDate	DateKey	Year	Month	Month Name	MMM	Month Year	Day	Day Name	Day
Friday, December 3, 2021	44533	2021	12	December	Dec	Dec 2021	3	Friday	
Friday, December 10, 2021	44540	2021	12	December	Dec	Dec 2021	10	Friday	
Friday, December 17, 2021	44547	2021	12	December	Dec	Dec 2021	17	Friday	
Friday, December 24, 2021	44554	2021	12	December	Dec	Dec 2021	24	Friday	
Friday, December 31, 2021	44561	2021	12	December	Dec	Dec 2021	31	Friday	
Friday, April 22, 2022	44673	2022	4	April	Apr	Apr 2022	22	Friday	
Friday, April 29, 2022	44680	2022	4	April	Apr	Apr 2022	29	Friday	
Friday, May 6, 2022	44687	2022	5	May	May	May 2022	6	Friday	
Friday, May 13, 2022	44694	2022	5	May	May	May 2022	13	Friday	
Friday, May 20, 2022	44701	2022	5	May	May	May 2022	20	Friday	
Friday, May 27, 2022	44708	2022	5	May	May	May 2022	27	Friday	
Friday, June 3, 2022	44715	2022	6	June	Jun	Jun 2022	3	Friday	
Friday, June 10, 2022	44722	2022	6	June	Jun	Jun 2022	10	Friday	
Friday, June 17, 2022	44729	2022	6	June	Jun	Jun 2022	17	Friday	
Friday, June 24, 2022	44736	2022	6	June	Jun	Jun 2022	24	Friday	
Friday, July 1, 2022	44743	2022	7	July	Jul	Jul 2022	1	Friday	
Friday, July 8, 2022	44750	2022	7	July	Jul	Jul 2022	8	Friday	
Friday, July 15, 2022	44757	2022	7	July	Jul	Jul 2022	15	Friday	
Friday, July 22, 2022	44764	2022	7	July	Jul	Jul 2022	22	Friday	
Friday, July 29, 2022	44771	2022	7	July	Jul	Jul 2022	29	Friday	
Friday, August 5, 2022	44778	2022	8	August	Aug	Aug 2022	5	Friday	
Friday, August 12, 2022	44785	2022	8	August	Aug	Aug 2022	12	Friday	
Friday, August 19, 2022	44792	2022	8	August	Aug	Aug 2022	19	Friday	
Friday, August 26, 2022	44799	2022	8	August	Aug	Aug 2022	26	Friday	
Friday, September 2, 2022	44806	2022	9	September	Sep	Sep 2022	2	Friday	
Friday, September 9, 2022	44813	2022	9	September	Sep	Sep 2022	9	Friday	
Friday, September 16, 2022	44820	2022	9	September	Sep	Sep 2022	16	Friday	
Friday, September 23, 2022	44827	2022	9	September	Sep	Sep 2022	23	Friday	
Friday, September 30, 2022	44834	2022	9	September	Sep	Sep 2022	30	Friday	
Friday, October 7, 2022	44841	2022	10	October	Oct	Oct 2022	7	Friday	
Friday, October 14, 2022	44848	2022	10	October	Oct	Oct 2022	14	Friday	
Friday, October 21, 2022	44855	2022	10	October	Oct	Oct 2022	21	Friday	
Friday, October 28, 2022	44862	2022	10	October	Oct	Oct 2022	28	Friday	
Friday, November 4, 2022	44869	2022	11	November	Nov	Nov 2022	4	Friday	
Friday, November 11, 2022	44876	2022	11	November	Nov	Nov 2022	11	Friday	
Friday, November 18, 2022	44883	2022	11	November	Nov	Nov 2022	18	Friday	
Friday, November 25, 2022	44890	2022	11	November	Nov	Nov 2022	25	Friday	
Friday, December 2, 2022	44897	2022	12	December	Dec	Dec 2022	2	Friday	
Friday, December 9, 2022	44904	2022	12	December	Dec	Dec 2022	9	Friday	
Friday, December 16, 2022	44911	2022	12	December	Dec	Dec 2022	16	Friday	
Friday, December 23, 2022	44918	2022	12	December	Dec	Dec 2022	23	Friday	
Friday, December 30, 2022	44925	2022	12	December	Dec	Dec 2022	30	Friday	
Friday, April 21, 2023	45037	2023	4	April	Apr	Apr 2023	21	Friday	
Friday, April 28, 2023	45044	2023	4	April	Apr	Apr 2023	28	Friday	
Friday, May 5, 2023	45051	2023	5	May	May	May 2023	5	Friday	
Friday, May 12, 2023	45058	2023	5	May	May	May 2023	12	Friday	
Friday, May 19, 2023	45065	2023	5	May	May	May 2023	19	Friday	
Friday, May 26, 2023	45072	2023	5	May	May	May 2023	26	Friday	
Friday, June 2, 2023	45079	2023	6	June	Jun	Jun 2023	2	Friday	
Friday, June 9, 2023	45086	2023	6	June	Jun	Jun 2023	9	Friday	
Friday, June 16, 2023	45093	2023	6	June	Jun	Jun 2023	16	Friday	
Friday, June 23, 2023	45100	2023	6	June	Jun	Jun 2023	23	Friday	
Friday, June 30, 2023	45107	2023	6	June	Jun	Jun 2023	30	Friday	
Friday, July 7, 2023	45114	2023	7	July	Jul	Jul 2023	7	Friday	

Week Number	IsYTD	IsMTD	IsWTD	IsCurrentYTD	IsCurrentMTD	IsCurrentW
49	0	1	0	0	0	0
50	0	1	0	0	0	0
51	0	1	0	0	0	0
52	0	0	0	0	0	0
1	0	0	0	0	0	0
17	0	0	0	0	0	0
18	0	0	0	0	0	0
19	0	1	0	0	0	0
20	0	1	0	0	0	0
21	0	0	0	0	0	0
22	0	0	0	0	0	0
23	0	1	0	0	0	0
24	0	1	0	0	0	0
25	0	1	0	0	0	0
26	0	0	0	0	0	0
27	0	1	0	0	0	0
28	0	1	0	0	0	0
29	0	1	0	0	0	0
30	0	0	0	0	0	0
31	0	0	0	0	0	0
32	0	1	0	0	0	0
33	0	1	0	0	0	0
34	0	0	0	0	0	0
35	0	0	0	0	0	0
36	0	1	0	0	0	0
37	0	1	0	0	0	0
38	0	1	0	0	0	0
39	0	0	0	0	0	0
40	0	0	0	0	0	0
41	0	1	0	0	0	0
42	0	1	0	0	0	0
43	0	0	0	0	0	0
44	0	0	0	0	0	0
45	0	1	0	0	0	0
46	0	1	0	0	0	0
47	0	0	0	0	0	0
48	0	0	0	0	0	0
49	0	1	0	0	0	0
50	0	1	0	0	0	0
51	0	1	0	0	0	0
52	0	0	0	0	0	0
0	0	0	0	0	0	0
16	0	0	0	0	0	0
17	0	0	0	0	0	0
18	0	1	0	0	0	0
19	0	1	0	0	0	0
20	0	0	0	0	0	0
21	0	0	0	0	0	0
22	0	1	0	0	0	0
23	0	1	0	0	0	0
24	0	1	0	0	0	0
25	0	0	0	0	0	0
26	0	0	0	0	0	0
27	0	1	0	0	0	0

External Tools: Date Table

Mark as Date Table

Mark as Date Table



Relationships

Options

GLOBAL

Data Load

Power Query Editor

DirectQuery

R scripting

Python scripting

Security

Privacy

Regional Settings

Updates

Usage Data

Diagnostics

Preview features

Auto recovery

Report settings

CURRENT FILE

Data Load

Regional Settings

Privacy

Auto recovery

Published dataset settings

Query reduction

Report settings

Type Detection

☐ Detect column types and headers for unstructured sources

Relationships

☒ Import relationships from data sources on first load ⓘ

☐ Update or delete relationships when refreshing data ⓘ

☐ Autodetect new relationships after data is loaded ⓘ

[Learn more](#)

Time intelligence

☐ Auto date/time ⓘ [Learn more](#)

Background Data

☒ Allow data previews to download in the background

Parallel loading of tables ⓘ

Maximum number of concurrent jobs [Learn more](#)

☒ Default

☐ One (disable parallel loading)

☐ Custom

Q&A

☐ Turn on Q&A to ask natural language questions about your data ⓘ [Learn more](#)

☐ Share your synonyms with everyone in your org

OK

Cancel

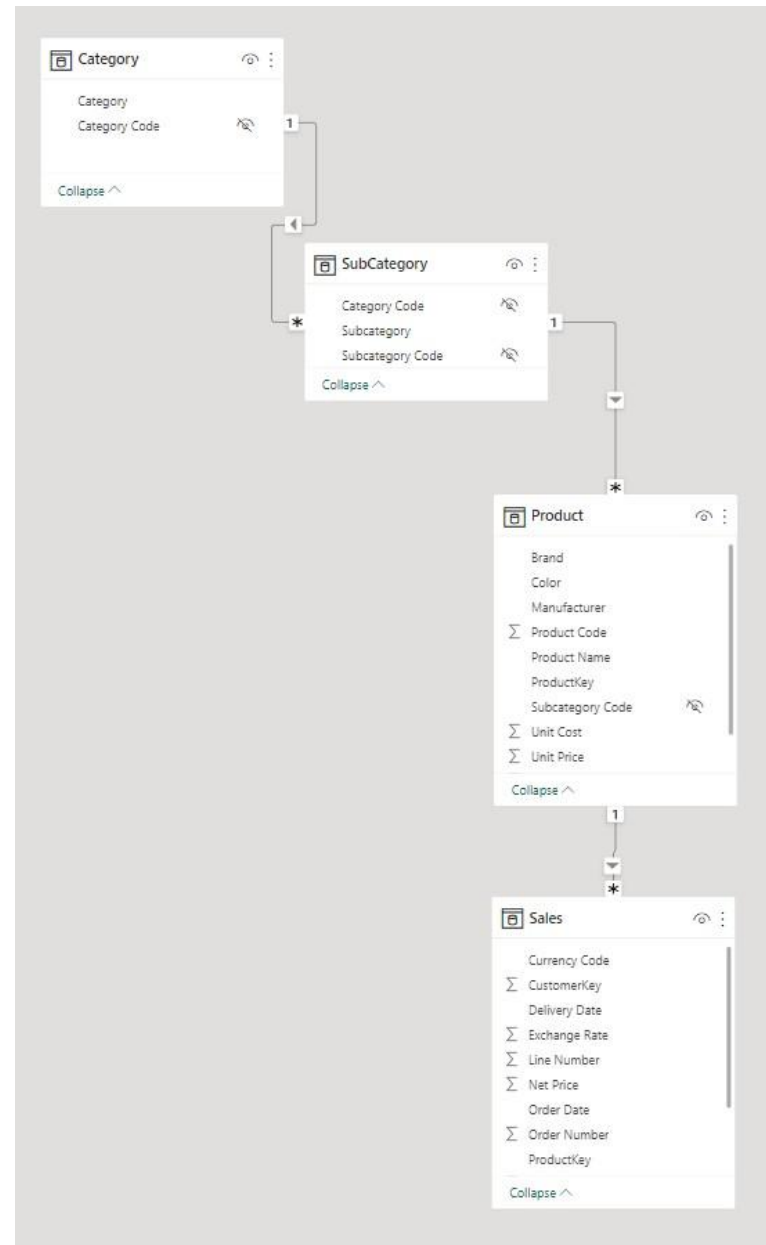
One big wide table (or tall)

SPSubCat_AddCusto...	🔍	⋮
Category.Category		
Σ Category.Category Code		
Currency Code		
Customer.Address		
Customer.Age		
Customer.Birthday		
Customer.City		
Customer.Continent		
Customer.Country		
Customer.Country Code		
Customer.CustomerKey		
Customer.Gender		
Customer.Name		
Customer.State		
Customer.State Code		
Customer.Zip Code		
CustomerKey		
Delivery Date		
Exchange Rate		
Line Number		
Net Price		
Order Date		
Order Number		
Product.Brand		
Σ Product.Category Code		
Product.Color		
Product.Manufacturer		
Product.Product Code		
Product.Product Name		
Product.ProductKey		
Σ Product.Subcategory Code		
Product.Unit Cost		
Product.Unit Price		
Product.Weight		
Product.Weight Unit Measure		
ProductKey		
Quantity		
StoreKey		
Collapse ^		

Wide Table

- Difficult to find the column you are looking for
- Refresh can take longer
- Performance is better with Star Schema
- DAX is more complicated

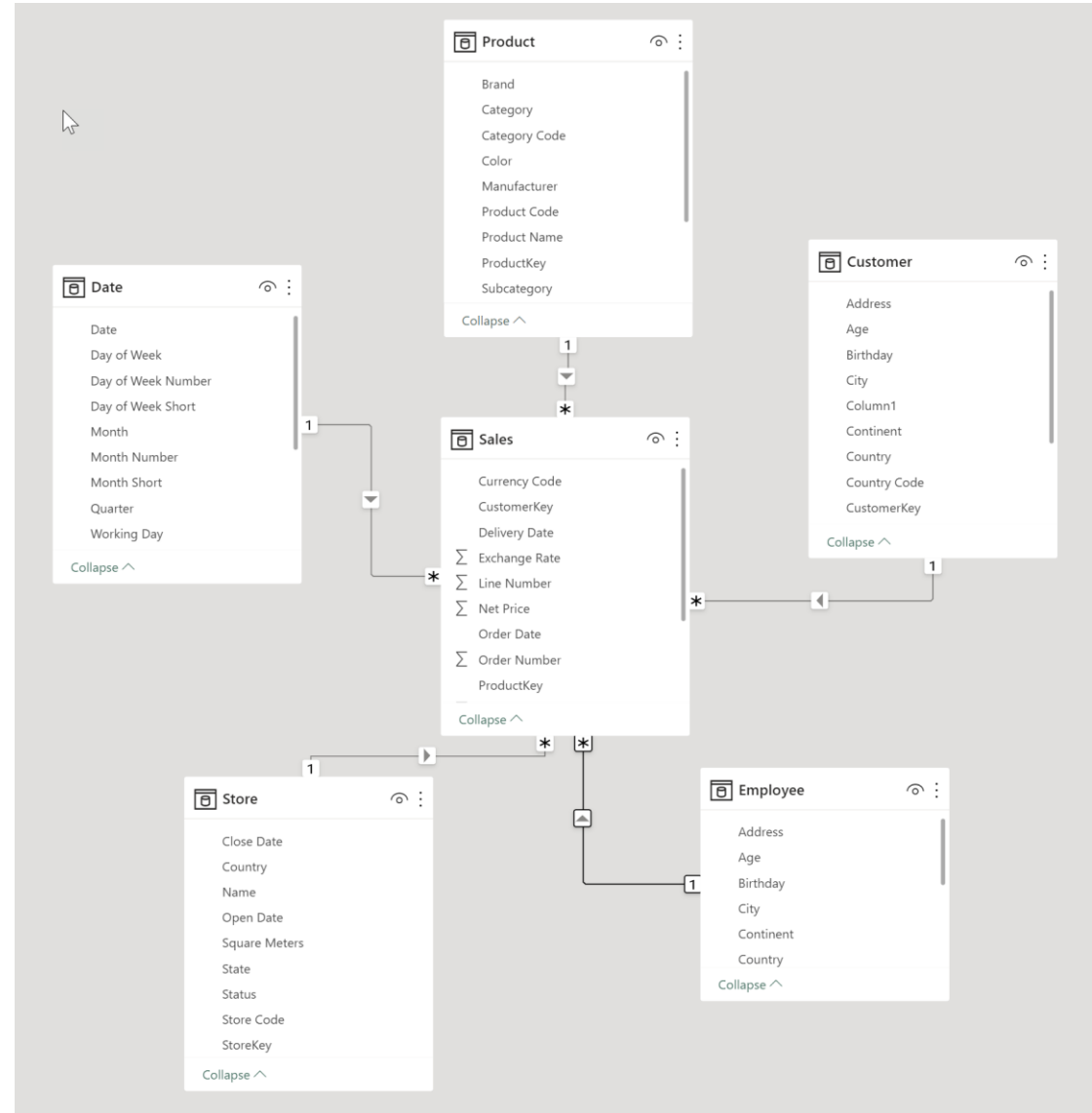
What is Snowflake Schema



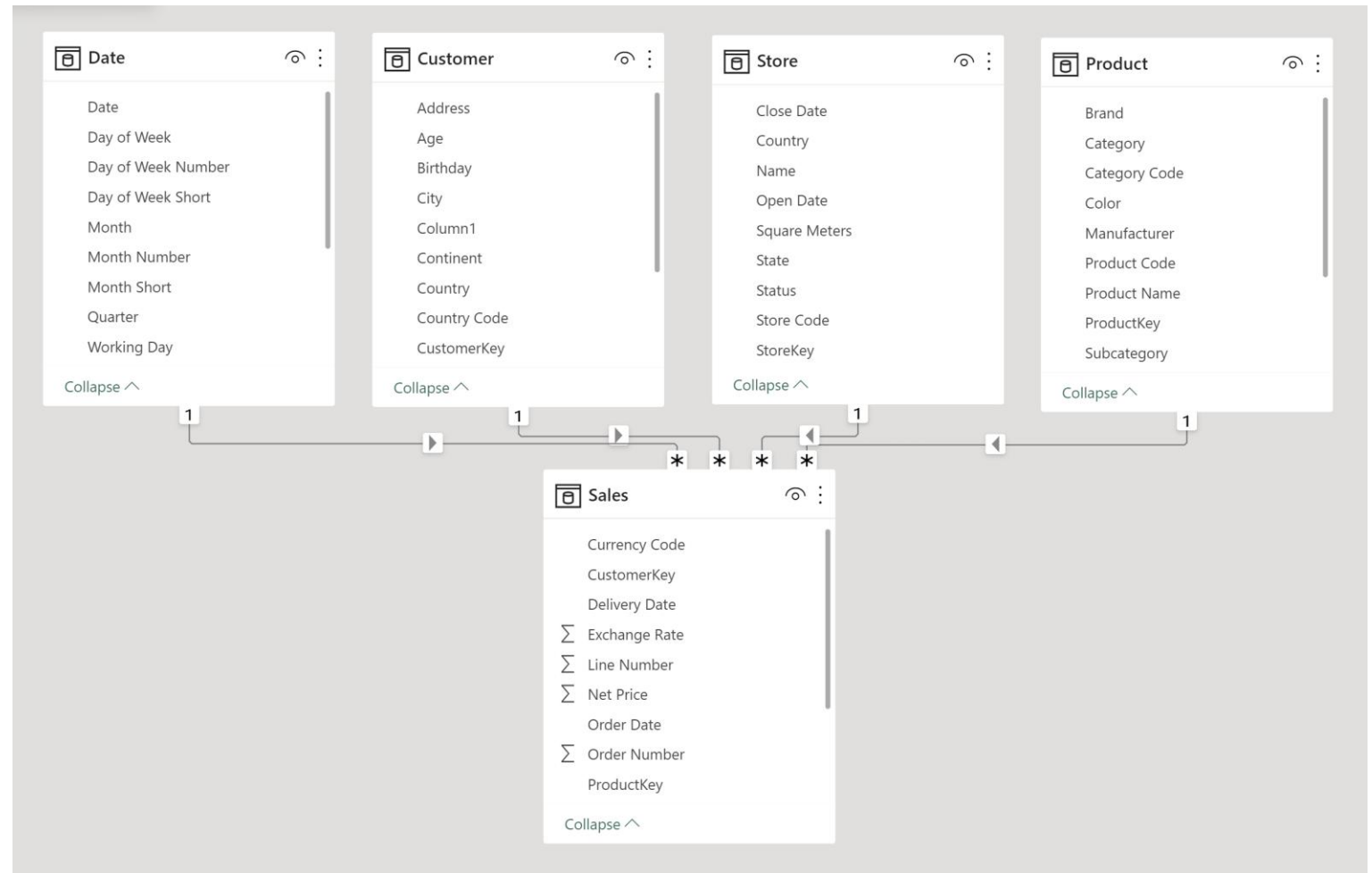
Other examples of Snowflake Schema

- General Ledger
 - Header and Detail
 - Combine into one table
- Employee
 - Employee SubCategory
 - Employee Category

What is a Star Schema



Collie Method



Snowflake versus Star Schema

Snowflake

- Harder to understand as a developer
- Expanded table issue with DAX

Star Schema

- Simpler
- Overall better performance

Expanded Tables

- When you apply a table filter it really is the expanded table.

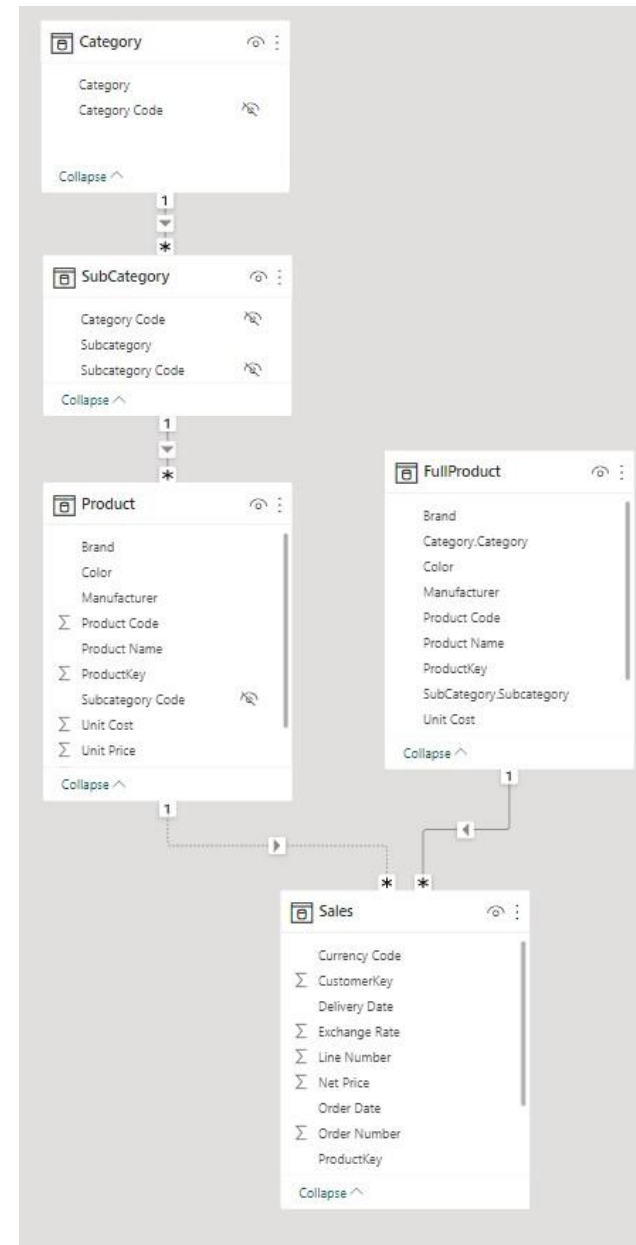
```
CALCULATE (  
    COUNTROWS (Subcategory),  
    Product  
)
```

- It is the expanded table for Product. So only those subcategories that have a product
 - Can't reference base table

DAX can be easier

- MAXX (
VALUES ('Product'[Brand]),  For each Brand
[Sales Amount]  Compute Max Sales Amount
)

Snowflake to (Super) Star Schema



Hiding Columns

- Hide Key Columns
- Build Measures on top of value columns and hide base column
- Implicit Measures versus Explicit Measures

Implicit Measures

- Currently if you create Calculation Groups it disables implicit measures
- Discourage Implicit Measures
 - True/False
- Build explicit Measures

Calculated Columns

- Create all calculated columns in source if possible
- Optimal sorting algorithm only on imported columns

Our Journey

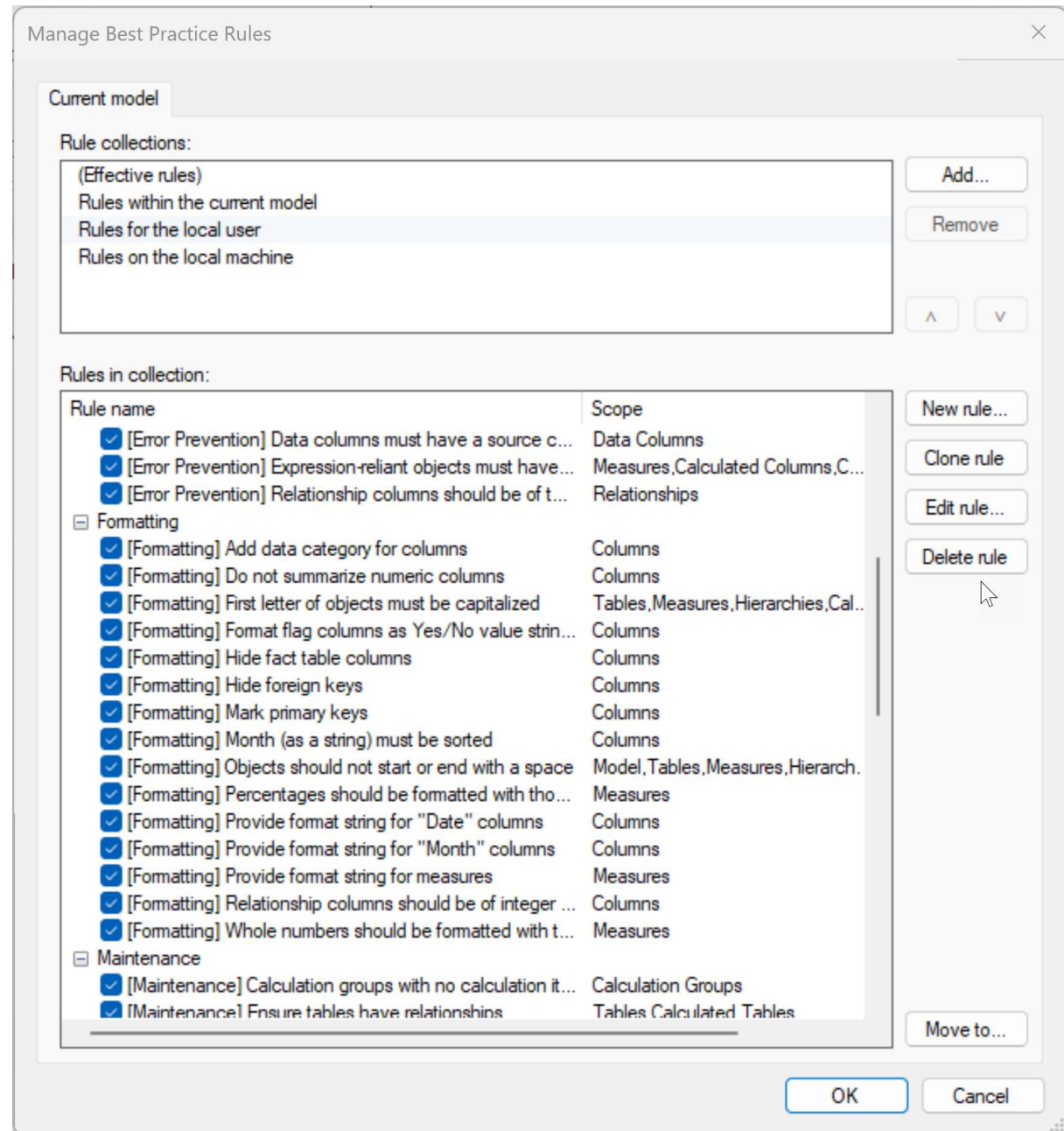


1. Intro
2. Power Query
3. Data Model
- 4. External Tools**
5. Conclusion

Tabular Editor

- With C# scripts you can automate creating or changing aspects of the model
 - For example, create a measure for every column that has the word amount

Tabular Editor – Best Practice Analyzer



Bravo: Date Table

Bravo does not support AutoDate



Unsupported

Models with auto date/time option enabled are not supported.

[Disabling auto date-time in Power BI \(video\)](#)

Time Intelligence

Options

GLOBAL

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CURRENT FILE

Data Load

Regional Settings

Privacy

Auto recovery

Published dataset settings

Query reduction

Report settings

Background Data

☐ Always allow data previews to download in the background

☒ Allow data previews to download in the background according to each file's setting

☐ Never allow data previews to download in the background

Parallel loading of tables

When you load data into Power BI (via import or DirectQuery), each data table is backed by a Power Query query. These queries are evaluated simultaneously instead of one-by-one, which can speed up the process. In certain situations, you might want to adjust the default number of simultaneous query evaluations and memory used. [Learn more](#)

Maximum number of simultaneous evaluations

20

Maximum memory used per simultaneous evaluation (MB)

432

Time intelligence

☐ Auto date/time for new files [Learn more](#)

Data Cache Management Options

Currently used: 655 KB

Clear Cache

Maximum allowed (MB):

4096

Restore Defaults

Q&A Cache Options

Currently used: 1.43 MB

Clear Cache

Maximum allowed (MB):

4096

OK

Cancel

Type of Calendar



Contoso10K > Manage Dates



Model Check



This model **is compatible with the Manage Dates feature.**

You can create new date tables without worrying about breaking measures or reports.

Calendar

Interval

Dates

Holidays

Time Intelligence

Choose a calendar template to apply to this model. Bravo will create the required tables or update them while keeping the existing relationships intact.



Template

Select *Monthly* for calendar based on different number of months. Set *Weekly* for 445-454-544-ISO calendars. Use *Custom* for flexible calendars of variable length. [Manage Templates](#)

Monthly - Fiscal ▾

First Month of the Year

For Weekly ISO, set *January*.

March ▾

Interval

Calendar Interval Dates Holidays Time Intelligence

Select a date interval for your model.



Date Interval

Choose how to determine the date interval of the model. Leave *First Year* and/or *Last Year* empty to use the automatic scan.

First Year

2018

Last Year

2024

Select Measures for Time Intelligence

StarSchema_Date > Manage Dates

Model Check



This model **is compatible with the Manage Dates feature.**

You can create new date tables without worrying about breaking measures or reports.

Calendar Interval Dates Holidays Time Intelligence

Create the most common Time Intelligence DAX functions available in your model.



Time Intelligence Functions



Target Measures

Choose the measure used to generate the Time Intelligence functions.

Choose Measures... ▾



Search



TABLE \ COLUMN



Sales

Time Intelligence Measures

<  StarSchema_Date > Manage Dates > Preview

Dates Time Intelligence



▼  To-date growth	A
▼  Sales Amount	A
 MOMTD % Sales Amount	A
 MOMTD Sales Amount	A
 PMTD Sales Amount	A
 PQTD Sales Amount	A
 PYTD Sales Amount	A
 QOQTD % Sales Amount	A
 QOQTD Sales Amount	A
 YOYTD % Sales Amount	A
 YOYTD Sales Amount	A
▼  To-date total	A
▼  Sales Amount	A
 MTD Sales Amount	A
 QTD Sales Amount	A
 YTD Sales Amount	A

Bravo: Optimize Model

Find unused columns, size of columns, and unique elements

Bravo Optimize Model

StarSchema

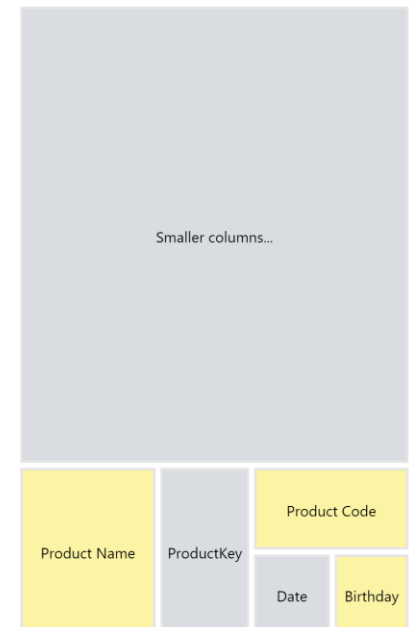
Your dataset is **1.48 MB** large and contains **66** columns, **56** of which are not referenced within the model.

Search Column

VPAX

	COLUMN	TABLE	CARDINALITY	SIZE	WEIG...
	Smaller columns...		10.44 K	1.08 MB	73%
	Product Name	Product	2.52 K	145.70 KB	10%
123	ProductKey	Product	2.52 K	97.86 KB	6%
	Product Code	Product	2.52 K	84.35 KB	6%
	Date	Date	1.10 K	41.05 KB	3%
	Birthday	Customer	511	40.34 KB	3%
			19.60 K	1.48 MB	100%

🔗 Unreferenced columns can generally be removed from the model to optimize performance. Before removing them, make sure you are not using these columns in any reports, which Bravo cannot determine.



Bravo
Save VPAX

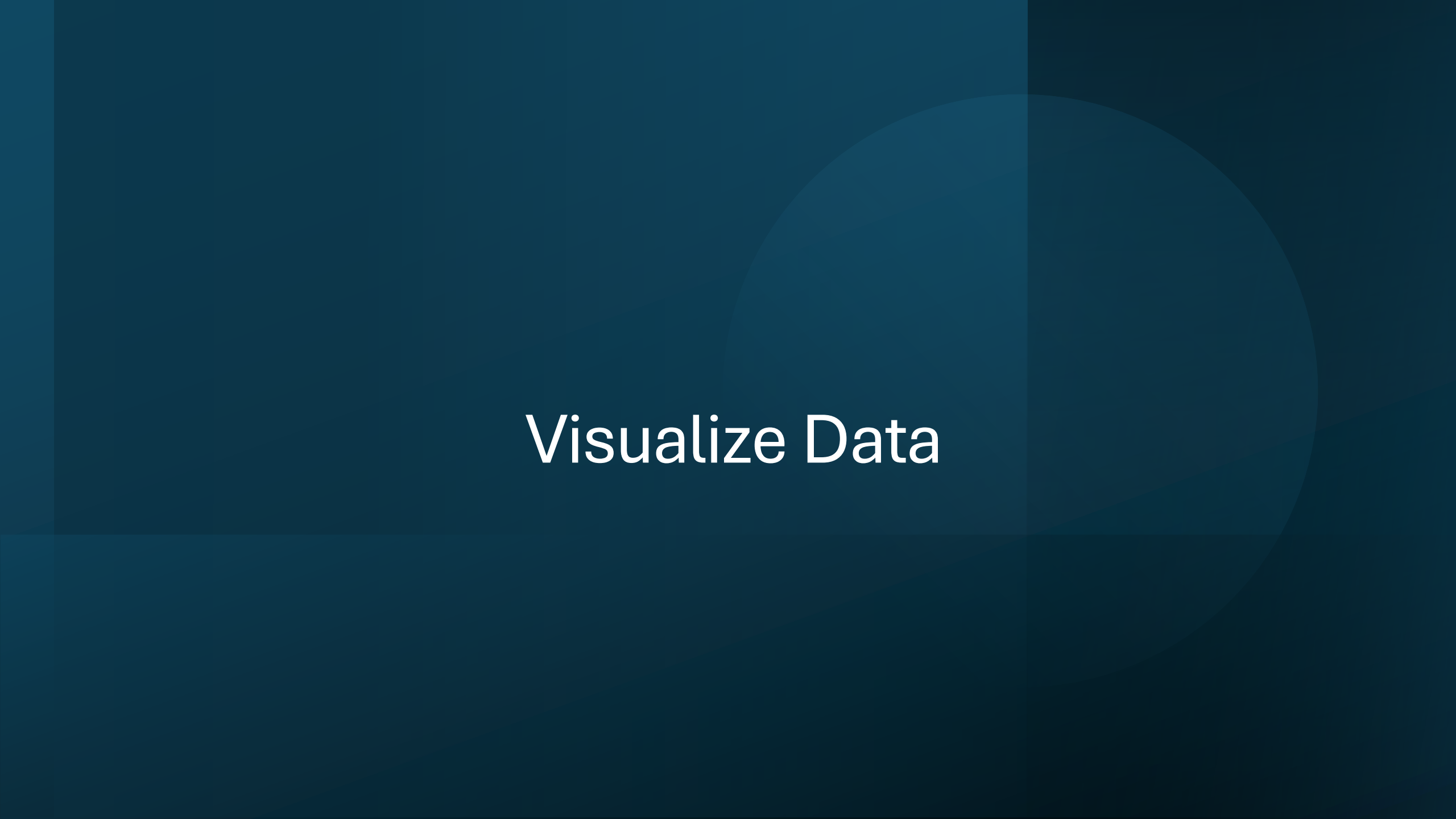
StarSchema

Your dataset is **1.48 MB** large and contains **66** columns, **56** of which are not referenced within the model.

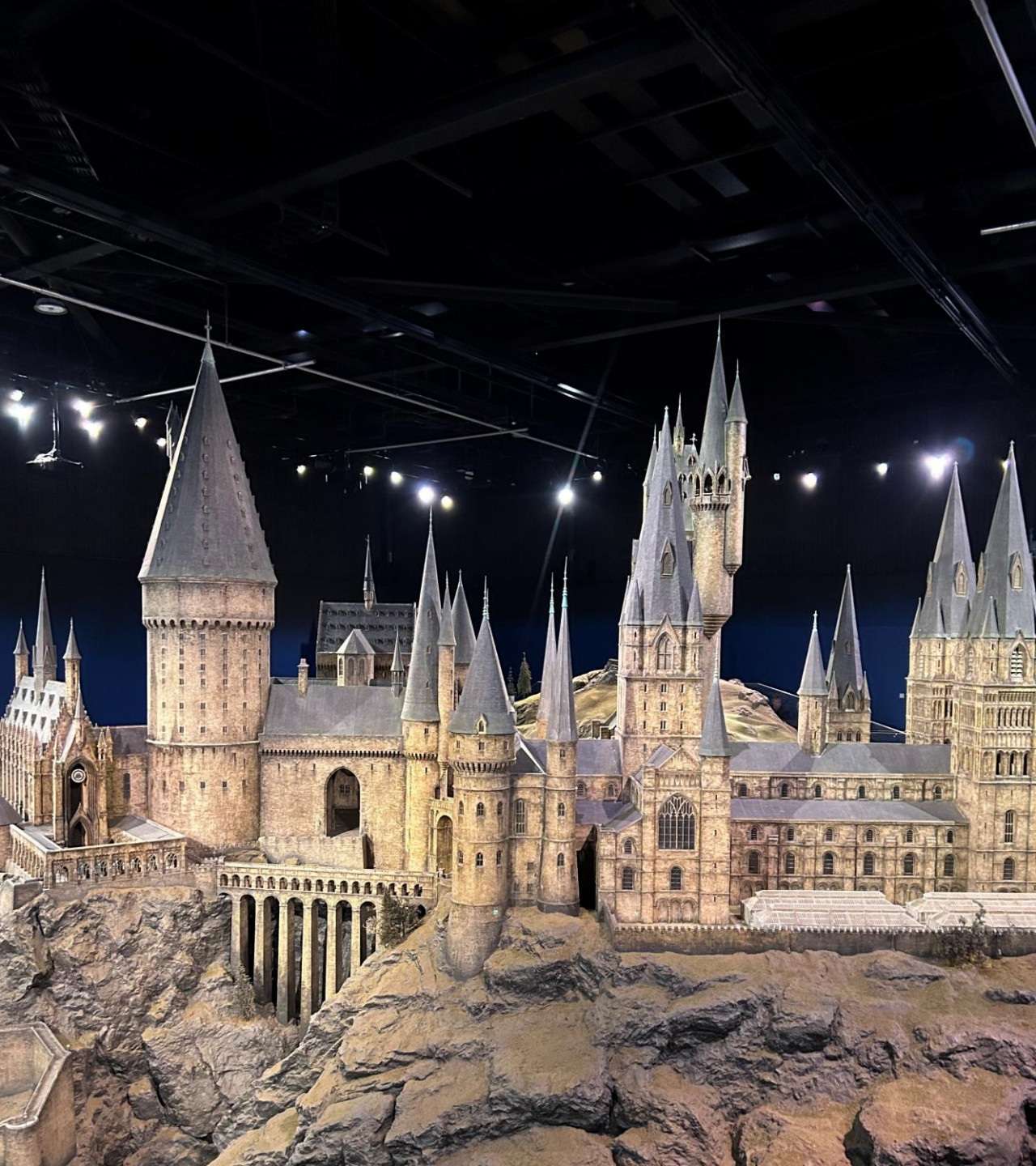


 VPAX

TABLE \ COLUMN	CARDINALITY	SIZE	WEIG...
>  Product	8.88 K	561.84 KB	37%
>  Customer	3.47 K	346.31 KB	23%



Visualize Data



Model Goals

Our Journey



1. Intro
2. Power Query
3. Data Model
4. External Tools
- 5. Conclusion**

Conclusion

- Shape data as close to source as possible
- Import it into a Star Schema
- Use external tools to help optimize model

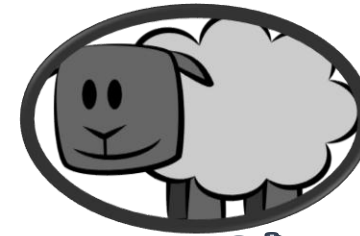
Resources

- [Guide to Query Folding](#)
- [Improve Model Performance](#)
- [Bravo for Power BI](#)
- [Tabular Editor 2](#)

Resources

- Kristyna's Date Table M
 - <https://github.com/Anytsirk12/DataOnWheels/tree/main/SQLBits%202024>
- Blog: Data on Wheels
 - <https://dataonwheels.wordpress.com/>
- DAX Studio
 - <https://daxstudio.org/>

Thank you



The Dax Shepherd

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